ORIGINAL ARTICLE

Cranial and cerebral signs in the diagnosis of spina bifida between 18 and 22 weeks of gestation: a German multicentre study

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ABSTRACT

Objective The aim of this article is to study secondary cranial signs in fetuses with spina bifida in a precisely defined screening period between 18 + 0 and 22 + 0 weeks of gestation.

Method On the basis of retrospective analysis of 627 fetuses with spina bifida, the value of indirect cranial and cerebral markers was assessed by well-trained ultrasonographers in 13 different prenatal centres in accordance with the ISUOG (International Society of Ultrasound in Obstetrics and Gynecology) guidelines on fetal neurosonography.

Results Open spina bifida was diagnosed in 98.9% of cases whereas 1.1% was closed spina bifida. Associated chromosomal abnormalities were found in 6.2%. The banana and lemon signs were evident in 97.1% and 88.6% of cases. Obliteration of the cisterna magna was seen in 96.7%. Cerebellar diameter, head circumference and biparietal diameter were below the 5th percentile in chromosomally normal fetuses in 72.5%, 69.7% and 52%, respectively. The width of the posterior horn of the lateral ventricle was above the 95th percentile in 57.7%. The secondary cranial and cerebral signs were dependent on fetal chromosome status and width of the posterior horn. Biparietal diameter was also dependent on the chromosome status with statistical significance $p = 0.0068$. Pregnancy was terminated in 89.6% of cases.

Conclusion In standard measuring planes, lemon sign, banana sign and an inability to image the cistern magna are very reliable indirect ultrasound markers of spina bifida. © 2014 John Wiley & Sons, Ltd.

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INTRODUCTION

Spina bifida is one of the most common malformations of the central nervous system. Prevalence depends on geographical, ethnic and seasonal influences and is between 0.2 and 0.6/1000 live births in Europe and North America.[1,2] Frequency can be significantly reduced by periconceptional folic acid supplementation.[2] There is also a connection with chromosomal anomalies, syndromes and teratogenic noxae.[2,3] Ultrasound diagnosis of spina bifida has substantially improved in the last 30 years because of improved image resolution in B scans and because secondary cranial signs have now been included. In 1975, Campell *et al.*[4] were the first to report on the prenatal diagnosis of three cases of lumbosacral spina bifida. At that time, ultrasound examinations were performed with compound ultrasound imaging (compound B scans). Because of increasing improvement in image resolution in modern ultrasound technology, it is possible today to detect spina bifida at second trimester screening in 88–96% of cases.[3,5–9] Ultrasound diagnosis, however, can be made more difficult because of unfavourable positioning. Indirect cranial and cerebral signs that indicate spina bifida have been reported since the 1980s.[10–13] These are changes in the shape of the head with bilateral scalloping of the frontal bones (the lemon sign), decreased head biometry measurements, increase in ventricular size, morphological changes in the cerebellum (the banana sign) with dorso-caudal displacement into the spinal canal resulting in obliteration of the cisterna magna (Chiari II malformation).[10–15] Moreover, these signs correlate relatively well with spina bifida, on the other opinions on their validity differ, which is probably caused by the different screening periods, poorly defined measuring planes and also improved ultrasound resolution.

Recent ultrasound studies between the 11 and 13 gestational weeks show that signs on caudal displacement of the posterior brain stem are present in fetuses with open spina bifida as early as the first trimester. This displacement causes compression of the fourth ventricle and/or cisterna magna, which subsequently leads to disappearance of what is called intracranial translucency.[16]

In Germany, the Joint Federal Committee (G-BA) amended the maternity guidelines in September 2010 (http://www.g-ba.de/downloads/62-492-773/RL_Mutter-2013-07-18_2013-09-20.pdf). As part of this revised version, pregnant women are offered an informed choice in the second trimester between the option of an ultrasound scan with biometry *without systematic* examination of fetal morphology and a scan with biometry *with systematic* examination of fetal morphology by a particularly well-qualified examiner. The systematic examination of fetal morphology includes the fetal head with the assessment of the ventricles, shape of the head and cerebellum as well as the neck and shoulder region with the assessment of dorsal skin contour.

The aim of this multicenter study is to review and re-evaluate, in a second ultrasound screening (18 + 0 to 22 + 0 weeks) by well-trained sonographers (DEGUM II and III) and modern ultrasound equipment, the importance of the indirect cranial and intracranial signs such as biparietal diameter (BPD), head circumference (HC), shape of the head cerebellum, ventricular width, and the ability to image the cistern magna using clearly defined benchmarks for the diagnosis of spina bifida.

METHODS

From January 2000 to June 2013, 627 fetuses with spina bifida between 18 + 0 and 22 + 0 weeks' gestation underwent retrospective analysis in 13 different prenatal centres in Germany (DEGUM stages II and III). The term DEGUM is an acronym standing for the Deutsche Gesellschaft für Ultraschall in der Medizin (German Society of Ultrasound in Medicine). The cases were found in the ViewPoint database (View Point 5.6.8.428; ViewPoint Bildverarbeitung GmbH, Weßling, Germany) and NEXUS database (SonoGeb, NEXUS-AG, Frankfurt, Germany). The ultrasound examinations were performed by specialised examiners only (DEGUM II and DEGUM III). Gestational age was calculated from the last menstrual period and confirmed by measurement of crown-rump length in the first trimester.[17] BPD and occipitofrontal diameter (OFD) were measured from 'outside to outside'.[18] HC was calculated using the formula $HC = 1.62 \times (BPD + OFD)$. The standard values proposed by Merz and Wellek and used in prenatal care in Germany were used as reference charts for BPD and HC.[19] When measured values for BPD and HC were below the 5th percentile, microcephaly was diagnosed.

Ultrasound assessment of the configuration of the frontal bones or evidence of scalloping of the frontal bones (the lemon sign) and the width of the posterior horn of the lateral ventricle (inside–inside) were made in the transventricular plane in accordance with the ISUOG guidelines[18] (Figure 1). The data obtained by Snijders *et al.*[20] were used as reference ranges for width of the ventricle. When the width of the posterior horn of the lateral ventricle was ≥10 mm, ventriculomegaly was diagnosed. The posterior fossa, as well as the cerebellum and cisterna magna, was assessed in the transcerebellar plane.[18] An outside–outside measurement of the cerebellum was made in the transverse plane.[18,20,21] The data obtained by Snijders *et al.*[20] were used as reference ranges for the cerebellum. The cisterna magna was classified as pathological if imaging was impossible or the measured value was less than 2 mm. The measured values for BPD, HCs, cerebellar diameter and width of the posterior horn of the lateral ventricle were plotted in the relevant reference charts and compared.

In addition, the effects of the localisation of the spina bifida (cervical, thoracic, lumbar and sacral) on BPD, HC, cerebellar diameter and width of ventricle were investigated. The fetuses underwent detailed ultrasound assessment for associated malformations and chromosomal anomalies, and amniocentesis was performed if there were indications of the former or at the request of the parents. Detailed psychosocial, genetic and neuropaediatric counselling was given to the parents after spina bifida was diagnosed.

The following high-end ultrasound systems were used: Combison 330 and 530D MT, (Kretz-Technik, Zipf), Voluson 730 Expert + Voluson E8 (GE Healthcare, Kretz Ultrasound, Zipf, Austria), Sonoace Accuvix, Hitachi EUB 6500, Siemens Elegra, Acuson Sequoia, General Electrics Logiq 9, HDI 5000 Sono CT and Philips IU22 (Philips medical systems, Netherland), Aloka Prosund SSD, Aplio XG and Aplio 500 (Toshiba Medical System Europe, Netherland).

STATISTICAL ANALYSIS

Data analyses were performed using the statistical software package SAS (Version 9.2, SAS Institute Inc., Cary, NC, USA). Statistical results were reported as mean, median, range and proportion. In order to analyse differences in continuous or ordinal variables, *t*-tests or Wilcoxon-Rank-Sum tests were conducted as appropriate. Differences in proportions were analysed by means of chi-squared tests or exact chi-squared tests according to expected cell frequencies. Differences were considered statistically significant if their p-value was <0.05.

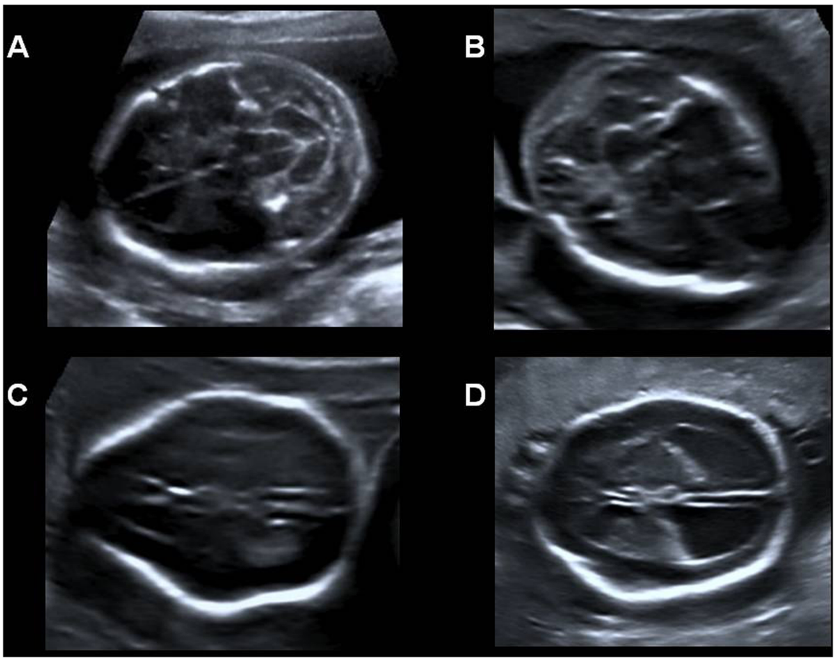

Figure 1 Fetus with lumbosacral spina bifida at 20 weeks of gestation: (A) banana sign, (B) obliteration of cisterna magna, (C) lemon sign and (D) ventriculomegaly

RESULTS

Mean maternal age for all women was 30.5 (range, 15–44) years. In women with fetuses with normal chromosomes, it was 30.4 (range, 15–44) years and 32.5 (range, 20–44) years in women with fetuses with abnormal chromosomes ($p = 0.018$). Median body mass index (BMI) was 25.2 (15.9–55.1) in all women, 25.4 (range, 15.9–55.1) in women with fetuses with normal chromosomes and 23.5 (range, 19.1–31.6) ($p = 0.013$) in women with fetuses with abnormal chromosomes. Prenatal diagnosis of spina bifida was made at a median gestational age of 20.9 (range, 17.7–22.9) weeks' gestation. The history of the pregnancy in relation to neural tube defects including use of anticonvulsants was negative except for two women. In six cases, type I diabetes was present and in eight cases gestational diabetes. The percentage of dichorionic diamniotic twin pregnancies where one fetus had spina bifida was 4% (25/627). Periconceptional folic acid prophylaxis had been given in 85.5% (536/627) of cases. Amniocentesis was performed if there was ultrasound evidence of associated malformations or at the request of the parents in 15.5% (97/627). The overall rate of chromosomal abnormalities was 6.2% (39/627) (trisomy 18 ($n = 21$), triploidia ($n = 11$), trisomy 13 ($n = 5$), Klinefelter syndrome ($n = 1$) and one case of tetrasomy 9p). Associated malformations were found in 4.6% (27/588) of the fetuses with normal chromosomes (heart defects ($n = 13$), OEIS complex ($n = 4$), renal anomalies ($n = 3$), gastroschisis ($n = 2$), congenital diaphragmatic hernia ($n = 2$), body stalk anomaly ($n = 1$), anal atresia ($n = 1$), and bladder exstrophy ($n = 1$)).

The banana sign and lemon sign were present in 97.1% (571/588) and 88.6% (521/588) of the spina bifida fetuses with normal chromosomes (Table 1). Obliteration of the cisterna magna was evident in 96.7% (569/588). Cerebellar diameter, HC and BPD below the 5th percentile were found in fetuses with normal chromosomes in 72.5% (338/466), 69.7% (410/588) and 52% (306/588) of cases, respectively. Width of the posterior horn of the lateral ventricle was above the 95th percentile in 57.7% (339/588), ≥10 mm in 46.1% (271/588) and ≥12 mm in 20.8% (122/288) (Figures 2–5).

Table 1 Comparison of different cranial and cerebral signs with the chromosomal status of spina bifida fetuses

Variable	Spina bifida fetuses with normal chromosomes % (n)	Spina bifida fetuses with abnormal chromosomes % (n)	p-value
BPD < 5%	52.0 (306/588)	74.4 (29/39)	0.0068
HC < 5%	69.7 (410/588)	82.1 (32/39)	ns
Vp ≥ 95%	57.7 (339/588)	30.8 (12/39)	0.0011
Vp ≥ 10 mm	46.1 (271/588)	28.2 (11/39)	0.0297
Lemon sign	88.6 (520/588)	74.4 (29/39)	0.0141
Banana sign	97.1 (571/588)	94.9 (37/39)	ns
Cerebellum <5%	72.5 (338/466)	62.1 (18/29)	ns

chi = exact chi-squared test.
BPD, biparietal diameter; HC, head circumference; Vp, posterior ventricle; ns, not significant.

In the fetuses with abnormal chromosomes, obliteration of the cisterna magna was detected in 94.9% (37/39). HC, BPD and cerebellar diameter were below the 5th percentile in 82.1% (32/39), 74.4% (29/39) and 62.1% (18/29) of cases, respectively. Width of the posterior horn of the lateral ventricle was above the 95th percentile in 30.8% (12/39), ≥10 mm in 28.2% (11/39) and ≥12 mm in 15.4% (6/39). The lemon sign and banana sign were present in 74.4 (29/39) and 94.9% (37/39) of the spina bifida fetuses with abnormal chromosomes (Table 1).

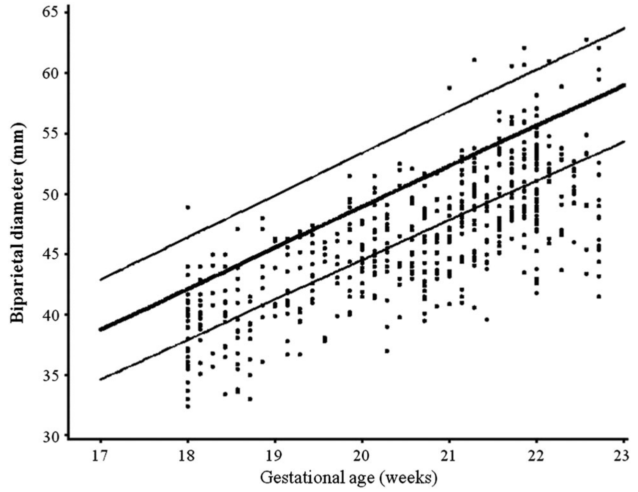

Figure 2 Individual measurements of biparietal diameter (BPD) in 588 chromosomal normal fetuses with spina bifida plotted in the modified reference ranges of Merz *et al.*[19] limited to the 17–23 weeks interval. Solid lines showing 5th, 50th and 95th percentile

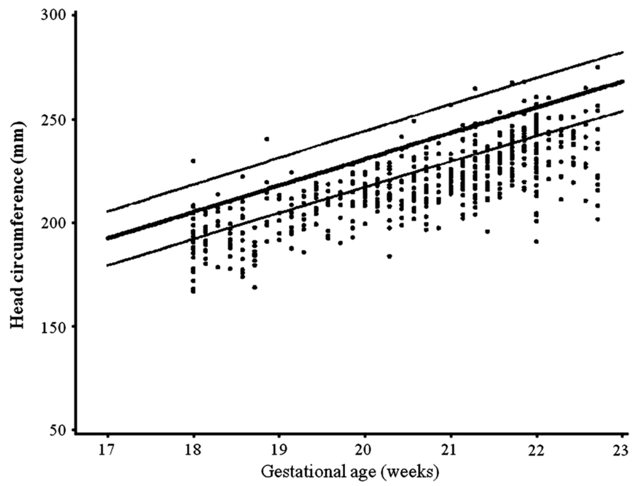

Figure 3 Individual measurements of head circumference (HC) in 588 chromosomal normal fetuses with spina bifida plotted in the modified reference ranges of Merz *et al.*[19] limited to the 17–23 weeks interval. Solid lines showing 5th, 50th and 95th percentile

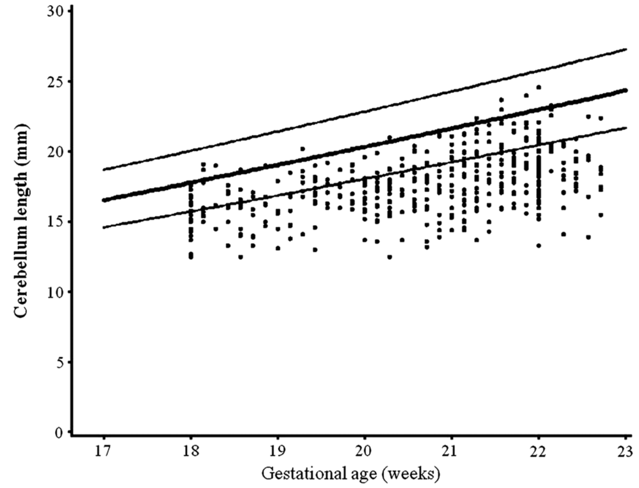

Figure 4 Individual measurements of length of the cerebellum in 466 chromosomal normal fetuses with spina bifida plotted in the modified reference ranges of Snijders *et al.*[20] limited to the 17–23 weeks interval. Solid lines showing 5th, 50th and 95th percentile

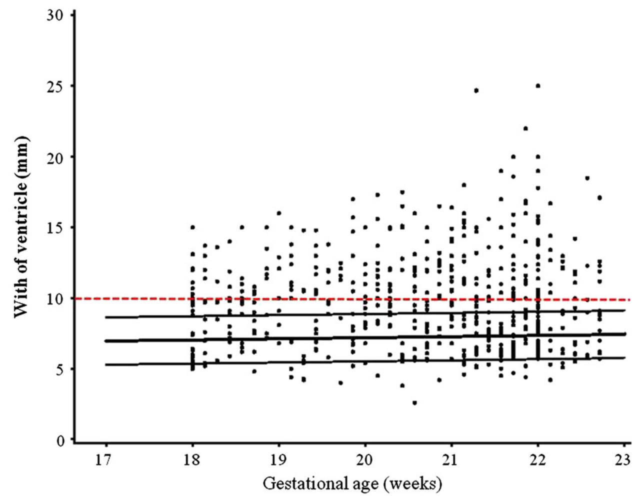

Figure 5 Individual measurements of the width of posterior ventricle in 588 chromosomal normal fetuses with spina bifida plotted in the modified reference ranges of Snijders *et al.*[20] limited to the 17–23 weeks interval. Solid lines are showing 5th, 50th and 95th percentile. Broken line represents the cut-off of 10 mm

When BPD in spina bifida fetuses with normal chromosomes was compared with that of spina bifida fetuses with abnormal chromosomes, a statistically significant difference was found (52% (306/588) vs 74.4% (29/39)); $p = 0.0068$. A statistically significant difference was similarly found with regard to ventricular width above the 95th percentile (57.7% (339/588) vs 30.8% (12/39)), $p = 0.0011$ (Table 1). Furthermore, the lemon sign was significantly less pronounced in fetuses with chromosomal anomalies than in healthy fetuses (74.4% vs 88.6%, $p = 0.0141$). When width of the posterior horn of the lateral ventricle (Vp) (<10 mm vs ≥10 mm) was compared in the fetuses with normal chromosomes, statistically significant differences emerged for BPD, HC and for detection of lemon and banana signs (Tables 2). When ventricular width was ≥10 mm, lower BPD and lower HC were significantly more common, and detection of the banana sign was significantly less likely. The lemon sign, in contrast, was more frequently detected.

When length of pregnancy 18 + 0–20 + 0 SSW ($n = 183$) and 20 + 1–22 + 0 SSW ($n = 335$) were compared, significant differences were found in fetuses with normal chromosomes in relation to BPD [(41.3% vs 55%), $p = 0.0023$] and HC [(61.8% vs 71.3%), $p = 0.025$].

The location of spina bifida was as follows: cervical in 0.3% of cases, thoracic in 7%, lumbar in 67.3% and sacral in 24.2% of the whole group.

The localisation of the spina bifida had no effect on BPD, HC, ventricular width and lemon and banana signs. In 98.9% (621/627) of cases, open spina bifida was diagnosed and in 1.1% (6/627) closed, whereas in 5 cases out of 6, no secondary cranial and cerebral signs were detected.

Table 2 Comparison of different cranial and cerebral signs with the width of the ventricles in chromosomal normal fetuses

Variable	Ventricle < 10 mm % (n)	Ventricle ≥ 10 mm % (n)	p
BPD < 5%	59.6 (189/317)	43.2 (117/271)	0.0001
HC < 5%	75.7 (240/317)	62.7 (170/271)	0.0006
Lemon sign	85.1 (269/316)	92.6 (251/271)	0.0044
Banana sign	95.3 (303/317)	99.3 (269/271)	0.0040
Cerebellum <5%	71.4 (177/248)	73.9 (161/218)	ns

chi = chi-squared test.
BPD, biparietal diameter; HC, head circumference.

The pregnancy was terminated in the singleton pregnancies and selective feticide performed in twin pregnancies in 85.5% (536/627) of cases. One late abortion occurred as a result of cervical incompetence. In addition, one intrauterine fetal death occurred in the third trimester. Intrauterine loss rate therefore was 3.1% (2/65).

DISCUSSION

The most common localisation for spina bifida is the lumbar and sacral region of the spine. There is generally a defect of the spinal cord, meninges, vertebrae and skin. The clinical picture varies depending on the severity of the defect. Open spina bifida accounted for 98.9% of our cases, which is consistent with the findings of other study groups.[22] Open neural tube defects cause changes in cranial and intracranial morphology induced by changes in cerebrospinal fluid (CSF) and CSF pressure. Downward displacement of the medulla oblongata, cerebellar tonsils, pons and fourth ventricle with obliteration of the cisterna magna results in Chiari II malformation.[23] This malformation is found in almost all myelomeningoceles with differing severity in the region of the neuroectoderm and surrounding mesoderm. Neuroectodermal changes include caudal elongation of the cerebellar tonsils and vermis and also of the caudal brainstem (medulla and pons) into the cervical spinal canal. In addition, the cerebellar hemispheres are deformed with obliteration of the posterior fossa.

Cerebellum (banana sign)

Caudal elongation of the cerebellar structures results in diminishment of the cisterna magna with changes in the shape of the cerebellum (banana sign) and is called a Chiari II malformation.[23] Whereas the lemon sign can frequently no longer be imaged after the 24th week of gestation, cerebellar changes and obliterations in the region of the cerebro-medullary fossa are also reliable indirect signs of open spina bifida even after this time. The incidence of the banana sign in the second trimester varies in the literature between 62% and 95%.[11,12,14] In agreement with the published data, we found a banana sign in 97.1% of cases and obliteration in the cisterna magna in 96.7%. The higher detection rates for the banana sign are primarily a result of the improved ultrasound technology in recent years. In sonographic assessment of the cerebellum in spina bifida, reduced transverse diameters are also found. Pilu *et al.*[21] reported that on 19 spina bifida fetuses, all of which had cerebellar diameter below the second standard deviation for gestational age. Similar results were reported by D'Addario *et al.*,[10] who found a reduced cerebellar diameter below the 10th percentile in 96% of cases. Our results indicate in 72.5% of cases reduced transverse cerebellar diameter below the 5th percentile. The differing results in the literature are predominantly based on the use of difference reference ranges and the thresholds of the latter.

Head shape (lemon sign)

Normally, the frontal bones on both sides appear in ultrasound imaging as an echogenic convex contour. In spina bifida fetuses, this shape is absent, and the shape of the skull is more likely to show inward scalloping, which is reminiscent of a lemon (known as the lemon sign).[24] The lemon sign is closely associated with spina bifida. Although the exact pathogenesis for the development of the lemon sign has not been fully clarified, it is suspected that, as a result of abnormal CSF flow and decompression, the collagen bundles that are positioned in the region of the frontal bones are drawn out to result in a concave deformity in the region of the two frontal bones.[23] Our study results show an incidence of 88.6% and are thus consistent with the data from the literature. Incidence of the lemon sign before the 24th week of gestation varies in the literature between 53% and 100% (Table 3).[10–12,14,15,25] The wide range of variation in the studies can probably be explained by the large time intervals between examinations (16–24 weeks).[15] Whereas Nicolaides *et al.*[11] found the lemon sign before the 24th week of gestation in 100% of cases, van den Hof *et al.*[12] detected the lemon sign in fetuses with open spina bifida before week 24 of gestation in 98% and after week 24 of gestation in only 13% of cases. Comparison of our two lengths of pregnancy (18 + 0–20 + 0 and 20 + 1–22 + 0 weeks) yielded no significant difference in the detection rate for the lemon sign. The lemon sign is not specific to spina bifida, however, but can also be detected in other structural and chromosomal abnormalities.[26] According to our results, the lemon sign is significantly less pronounced in fetuses with abnormal chromosomes than in spina bifida fetuses with normal chromosomes. In a normal group, a lemon sign is found in 0.7–1.3%.[11,12,14] An incorrect sonographic plane can also influence the incidence of the lemon sign and thus explain the different detection rates.[24]

Head size

Reduced BPD and HC below the 5th percentile are accompanied by microcephaly and appear to be due to loss of CSF in the region of the spinal defect.[23] In our study group of spina bifida fetuses with normal chromosomes, we detected reduced BPD and reduced HC below the 5th percentile in 52% and 69.7% of cases, respectively. Other authors have also reported reduced head growth but with a range of variation from 16% to 69% (Table 3).[11,13,25] In the study by Nicolaides *et al.*,[11] 70 fetuses with open spina bifida were examined at between 16 and 23 weeks' gestation. BPD und HC below the 5th percentile were found in 61% and 26% of cases, respectively.[11] In contrast to the study by Nicolaides *et al.*,[11] a

2.7-fold higher rate of reduced HC was found in our results, particularly when HCs were compared. This may be explained by the substantial differences in case numbers in the relevant studies on the one hand and the use of different reference charts on the other. Thiagarajah *et al.*[25] also detected a reduced BPD below the 5th percentile in 69% of cases in 16 fetuses between 16 and 24 weeks' gestation. Wald *et al.*,[13] however, found reduced head growth in 16% of cases only in a study of 20 fetuses with spina bifida in the second trimester. In a study that also involved small case numbers, Karl *et al.*[27] and Khalil *et al.*[28] found BPD to be below the 5th percentile between 16 and 24 weeks' gestation in 69.5% and 56%, respectively (Table 3). Assessment of BPD in spina bifida fetuses appears to be dependent on gestational age. We demonstrated that when the gestational periods of 18 + 0–20 + 0 and 20 + 1–22 + 0 weeks' gestation are compared, a significant difference exists both for BPD (41.3% vs 55%) and for HC (61.8% vs 71.3%). This observation is supported by current studies of BPD measurements in spina bifida fetuses in the first trimester, in whom decreased BPD values below the 5th percentile were found in 26–50% of cases only.[27–29] The gestational age at the time of measurement would thus appear to have an effect on the prevalence of reduced BPD and HCs in spina bifida fetuses. Besides the different size of the study groups, the differing information on incidence for reduced BPD may be explained by the use of different reference charts.[19,20,30] In a comparison of different reference ranges, Kurmanavicius *et al.*[30] showed, however, that the reference charts that we used for BPD and HC had no significant variations particularly in the region of the 5th percentile during the study time frame of 18 + 0–22 + 0 weeks' gestation. Standardised BPD and HC measurements thus show high reproducibility in the second trimester.

Our results further show that reduced BPD and HCs are associated with the width of the posterior horn of the lateral ventricle and the chromosome status of the spina bifida fetuses. In the second trimester, decreased BPD (59.6 vs 43.2%) and HCs (75.7% vs 67.7%) are significantly more common both when width of ventricle is ≥10 mm and when chromosome anomaly is detected (52% vs 74.4%). Genetic and syndromal anomalies that are associated with an increased rate of microcephaly and possibly CSF circulation disorders that have an effect on reduced head growth[23] may be the cause of this. Ultimately, the cause for this is unclear.

Khalil *et al.*[28] described in a multiple regression analysis a connection between decreased BPD and increased BMI, whereas Bernard *et al.*[29] were unable to detect any dependence of BPD on increased BMI and smoking. The discrepancies can most likely be explained by the differences in study population as well as the low case numbers. Thus, in the studies conducted by Khalil *et al.*,[31] spina bifida fetuses with chromosome anomalies, multiple pregnancies and structural anomalies were excluded, whereas Bernard *et al.*[29] included these cases. Our study results indicate that spina bifida fetuses with chromosomal anomalies have significantly lower BPD and HCs than euploid fetuses. It would appear that there is a connection between chromosomal anomalies and BPD. Furthermore, our results show a significantly higher BMI in mothers with a fetus with normal chromosomes compared with mothers with a spina bifida fetus with abnormal chromosomes.

Ventriculomegaly

Enlargement of the posterior horn of the lateral ventricle can also be a sign of spina bifida. The pathophysiological mechanism that results in the intrauterine development of ventriculomegaly and hydrocephalus arises primarily from caudal elongation of the brainstem. This gives rise to aqueduct stenosis, obstruction of the fourth ventricle and obliteration of the subarachnoid space, which result in abnormal CSF flow.[23] To measure ventricular width, both the ratio of the width of the posterior horn of the lateral ventricle (Vp) to the cerebral hemisphere (H) (cerebral hemisphere) (Vp/H) and more recently the straightforward inside–inside measurement of the posterior horn of the lateral ventricle are utilised.[11,12,32] The differing sonographic

Table 3 **Comparison of the detection rate of secondary cranial and cerebral marker in second trimester fetuses with spina bifida**

References	Period of study (weeks)	*n*	BPD < 5 percentile (%)	HC < 5 percentile (%)	Banana sign (%)	Lemon sign (%)	Ventriculomegaly >10 mm (%)	Obliteration cisterna magna (%)
Nicolaides *et al.* (*1986*)	16–23	70	61	26	95	100	86	95
Campbell *et al.* (*1987*)	16–23	26	62	35	62	100	54	
Nyberg *et al.* (*1988*)	<24	27				89	82	
Thiagarajah *et al.* (*1990*)	16–24	16	69			100	63	100
Van den Hof *et al.* (*1990*)	<24	107			72	98		96
Ghi *et al.* (*2006*)	16–34	53					64.2	
D'Addario *et al.* (*2008*)	18–28	49				53	81	96
Karl *et al.* (*2012*)	16–24	23	69					
Khalil *et al.* (*2013,2014*)	20–22	39	56				64.1	
Present study	18–22	588	52	69.7	97.1	88.6	46.1	96.7

BPD, biparietal diameter; HC, head circumference.

methods for assessing the posterior ventricle also explain the broad range of variation in the incidence of ventriculomegaly, which varies between 86% and 54% in the second trimester.[10,11,14,15,25] More recent publications refer to the significance of a standard imaging and measurement of the fetal intracranial ventricles (Table 3).[18,32] Nicolaides *et al.*[11] report on the detection of ventriculomegaly in the anterior and posterior horns of the lateral ventricle in 77% and 86% of cases, respectively, which is defined by Vp/H ratio as being above the 95th percentile. Using the assessment of posterior ventricular width, we found ventriculomegaly between 18 + 0 and 22 + 0 weeks' gestation below the 10th percentile in 57.7% and with a cut-off value of ≥10 mm in 44.9%. This rate is lower compared with the results obtained by Khalil *et al.*[31] and Thiagarajah *et al.*[25] who report an incidence of 64.1% and 63%, respectively. Sadly, we were unable to determine the Vp/H ratio, because the cerebral hemisphere was not measured. In a publication by Khalil *et al.*[31] that appeared recently, it was demonstrated that postnatal insertion of a ventriculoperitoneal shunt was always necessary when width of ventricle was ≥12 mm and Vp/H ratio ≥ 0.6 in the second trimester. This observation is important for the prenatal counselling of parents, because the cognitive and neurological development of the children is related to shunt complications, such as dislocation, infections and haemorrhage. However, hydrocephalus that requires a shunt may develop postnatally even when prenatal ventricular width was normal, which means that prognostic prenatal assessment ultimately remains problematic with regard to neurodevelopmental outcome. The localisation of the spina bifida has no effect, according to our results, on the width of the posterior horn of the lateral ventricle.

Chromosome anomalies

Although spina bifida is an isolated malformation in most cases, associated chromosomal anomalies are present in 2.6–9.7%.[2,3,33,34] In our study, we found chromosomal aberrations in 6.2% of cases and associated malformations in 4.6%. In this connection, trisomy 18 and the triploidies were the most common aneuploidies found, and of structural malformations, heart defects were the most common. These findings are consistent with the results from the literature.[3,33,34]

Spina bifida occulta

Prenatal diagnosis of occult spina bifida is difficult, however, particularly because no sufficiently clearly defined ultrasound criteria have been described because it rarely occurs and study groups to date are small. The incidence of closed spina bifida is given as between 4% and 7%.[10,22] In our group, we detected closed spina bifida in 1.1% of cases. In five out of six cases of occult spina bifida, we were unable to detect any structural abnormalities in brain anatomy. This observation is consistent with two cases reported by D'Addario *et al.*[10] and four cases described by Ghi *et al.*,[22] who also found no typical cranial and cerebral secondary signs of spina bifida occulta. Detection of ventriculomegaly, lemon sign und banana sign could thus be used to differentiate between open and closed spina bifida.[22]

Termination of Pregnancy (TOP)

In a large European multicentre study, a broad range of variation in termination of pregnancy (TOP) rate is reported when spina bifida is detected of between 17% and 100%.[5] A recently published meta-analysis reported a TOP rate of 63% (range, 31–97%).[35] The wide range for the TOP can be explained by many influences such as interdisciplinary prenatal counselling, the time of prenatal diagnosis, religious and ethical attitudes and the socio-economic environment of the parents. In our study, the parents decided on TOP in 89.6% of cases, which is consistent with the termination rates reported by D'Addario *et al.*[10] and Ghi *et al.*[22] (88% and 89.4%).

In Germany, counselling is given in accordance with the provisions of the German Gene Diagnostics Act (http://www.gesetze-im-internet.de/bundesrecht/gendg/gesamt.pdf), which includes the provision that the parents should be offered additional psychosocial human genetic and, in the case of spina bifida, neuropaediatric/neurosurgical counselling besides prenatal medical counselling. The parents themselves decide whether or not they wish to take up the additional counselling offered. The earliest TOP can take place is after the parents have had a period of at least 3 days to think it over after being informed of the diagnosis.

It can be stated in conclusion that, despite the retrospective analysis, this is currently the largest study to use established reference charts, the latest ultrasound technology with high resolution and standard measuring planes in relation to the incidence of cranial and cerebral secondary signs for detection of spina bifida. Reduced HC, abnormal head shape with scalloping of the frontal bone structure, known as the lemon sign, caudal elongation into the spinal canal and bow-shaped deformity of the cerebellum, known as the 'banana sign', inability to image the cisterna magna and ventriculomegaly particularly in the posterior ventricle are important secondary signs for spina bifida.

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WHAT'S ALREADY KNOWN ABOUT THIS TOPIC?

- Spina bifida is one of the most common malformations of the central nervous system. Certain signs during examination of fetal head with assessment of cerebellum, shape of the head, ventricles, BPD and head circumference correlate relatively well with spina bifida. Opinions on their validity differ.

WHAT DOES THIS STUDY ADD?

- The aim of this multicenter study is to review, in a second trimester ultrasound screening between 18 + 0 and 22 + 0 weeks of gestation, the value of the indirect cranial and intracranial signs such as cerebellum, shape of the head, ventricular width, biparietal diameter, head circumference and ability to image the cistern magna using clearly defined benchmarks, modern ultrasound equipment and well-trained sonographers for the diagnosis of spina bifida.

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